

The qualified instrument emerged through falsifiable gates
Failure / insufficient evidence, calibration, and replication remain distinct

METHOD CHANGE

One-shot L17
plus / minus / random

Duration isolation
one-shot vs sustained

Source/readout separation
paired-mean L27

Fresh full-dose
replication

Label-free dose
calibration

Fresh selected-dose
DEVELOPMENT test



FROZEN RESULT

No detectable
semantic movement

Duration contrast;
movement below threshold

Strong DEVELOPMENT signal;
historical safety issue

Movement replicated;
validity cost

D75 selected; no
accuracy/G/C/D selection

Strong safe DEVELOPMENT
replication

MICRO_Q1_NO_DETECTABLE
SIGNAL
N=50

GATE5_DURATION_EFFECT
BELOW_MOVEMENT
THRESHOLD
N=60

GATE6_3_SINGLE_MEAN
DESTRUCTIVE
N=60

GATE7_DESTRUCTIVE
N=120

GATE8_SAFE_LOWER_DOSE
SELECTED
N=50

GATE9_STRONG_SAFE
SELECTED_DOSE
REPLICATION
N=100

Gate 4

Gate 5

Gates 6-6.3

Gate 7

Gate 8

Gate 9